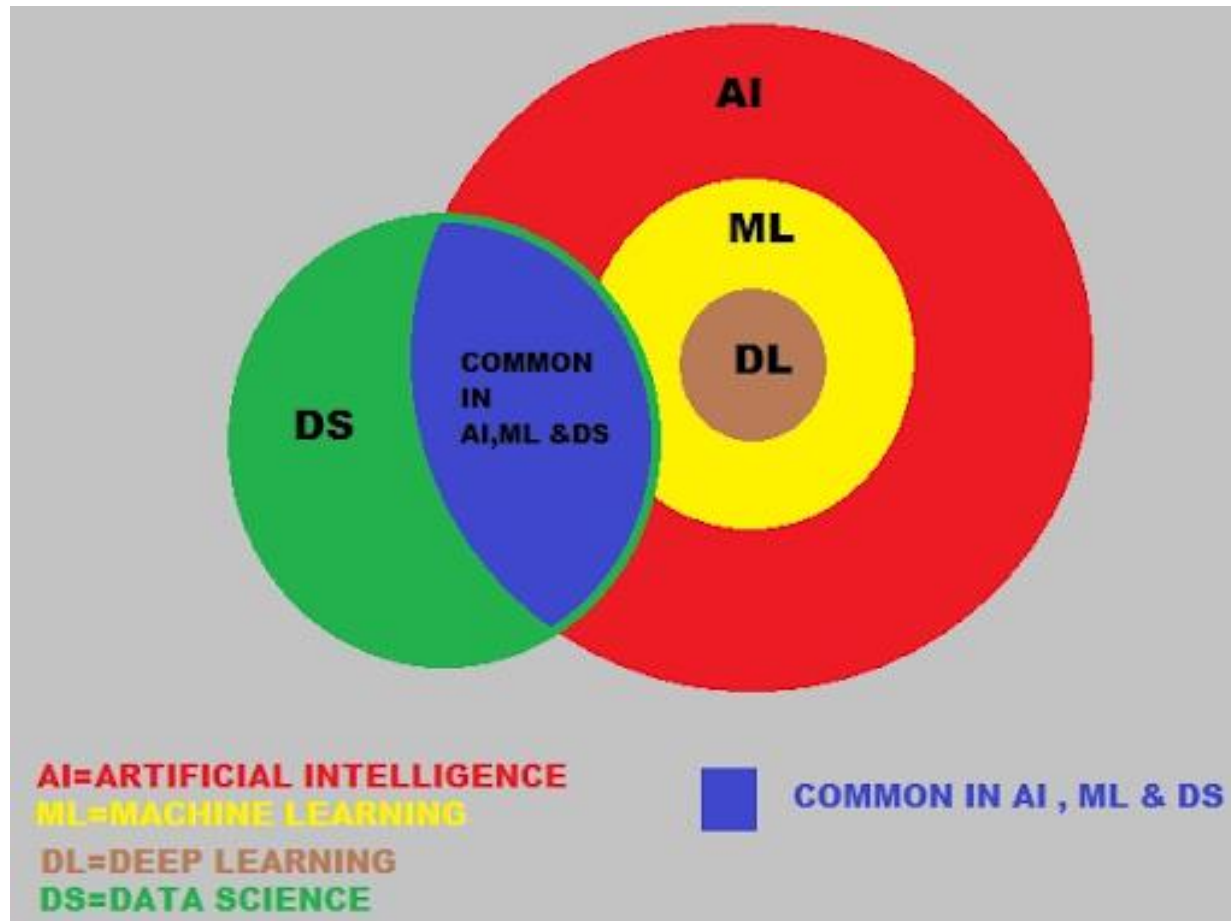


Introduction To Machine Learning

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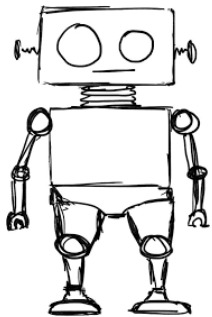
AI vs. ML vs. DL vs. DS



What is Machine Learning?



Go get the Pizza!!!



- Turn right
- Go 10 steps
- Turn left
- Go 5 steps
- Get the Pizza!!!

- **Complicated**
- **Not Generalized**

What is Machine Learning?



- Calculate distance to pizza.
- Move around & pick a direction where distance decreases.
- Take a step in that direction.
- Repeat until Pizza is found!!!



What is Machine Learning?

Learn From Experience



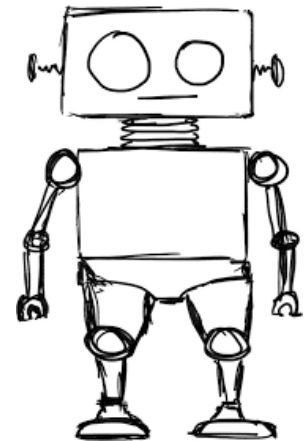
Human

Data
Learn from ~~Experience~~



Intelligent Machine

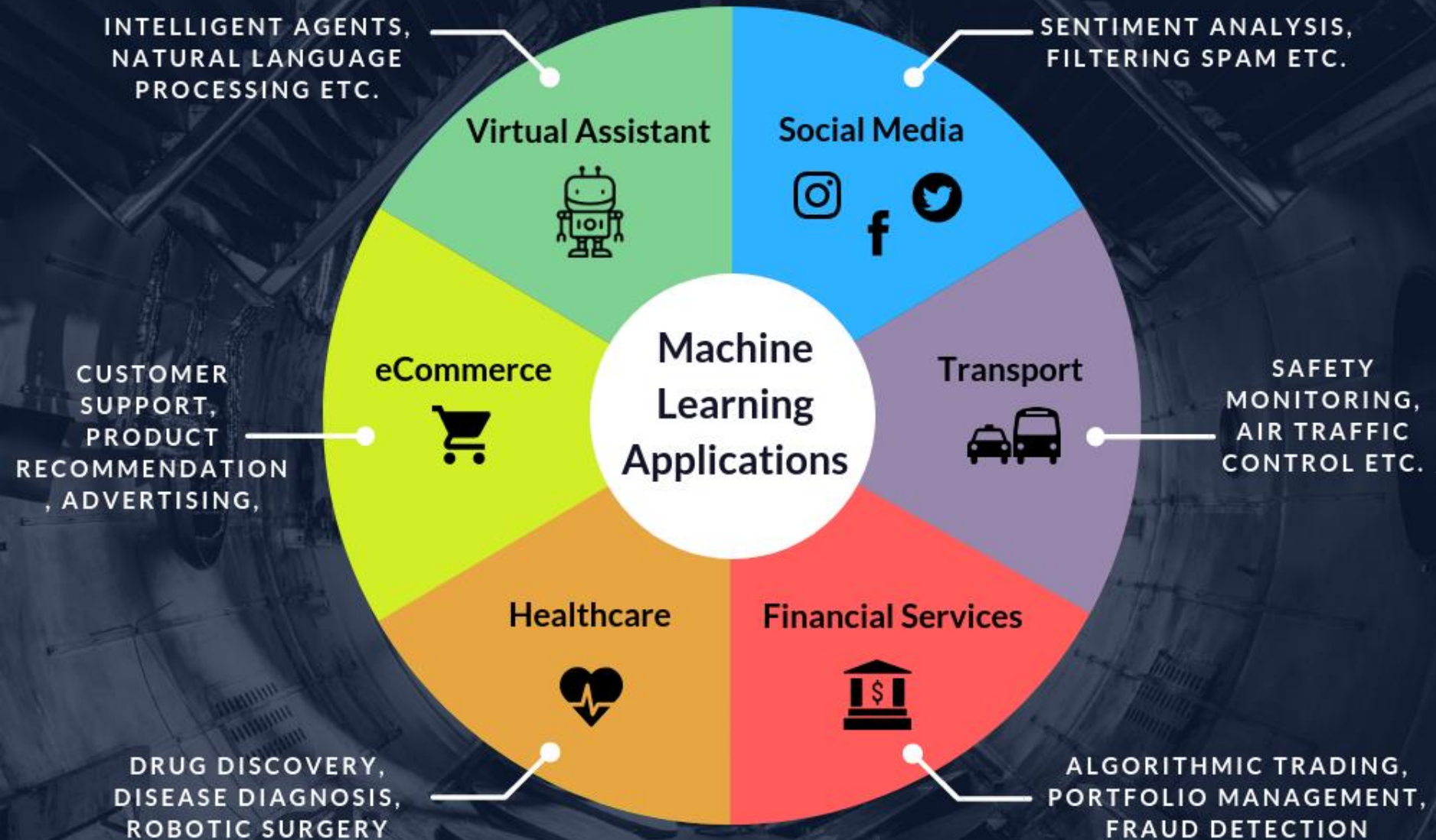
Follow Instructions



Machine

- Machine Learning is a subset of artificial intelligence where a computer/machine learns from the past experiences and makes future predictions.
 - The past experience is developed through the data collected.
 - In order to perform the task, the system learns from the data-set provided.
 - A data-set is a collection of many examples.
- **Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.**

APPLICATIONS OF MACHINE LEARNING



Applications of Machine Learning

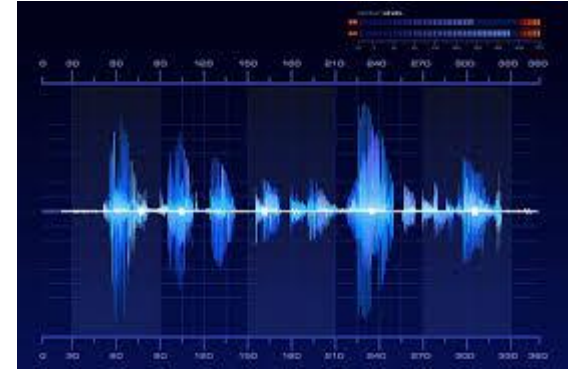
Medical images



Medical



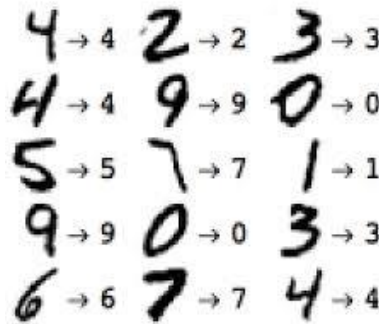
Robotics



Speech Recognition



Gaming



Handwriting Recognition



Natural Language Processing



Finance



Vision Systems

“A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.”

- Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam.
- What is the task T, experience E and performance measure P, in this setting?
- Classifying emails as spam or not spam. 
- Watching you label emails as spam or not spam. 
- The number (or percentage) of emails correctly classified as spam/not spam. 

Example of Machine Learning



100 sq-ft = 20 Lakhs

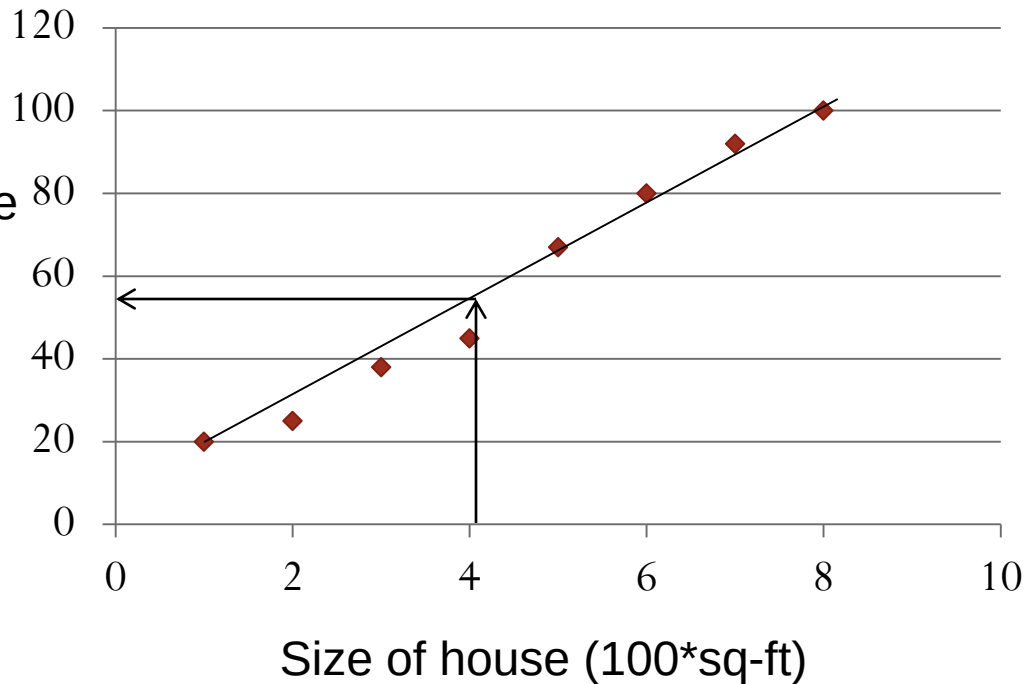


400 sq-ft = ?



600 sq-ft = 80 Lakhs

Price of house
(Lakhs)



Question:
What is the best
estimate for the
price of medium
house (400 sq-
ft)?

**Linear
Regression**

- **Gradient Descent**
- **Error Minimization**

Machine Learning Flow

(Steps used in Machine Learning)

- Data Collection and Preparation
- Feature Extraction
- Data Preprocessing
- Feature Selection
- Training-Validation-Testing
- Model Training
- Model Testing
- Model Evaluation
- Model Deployment

Data Collection and Preparation

- **Iris Dataset:**



Iris Versicolor



Iris Setosa

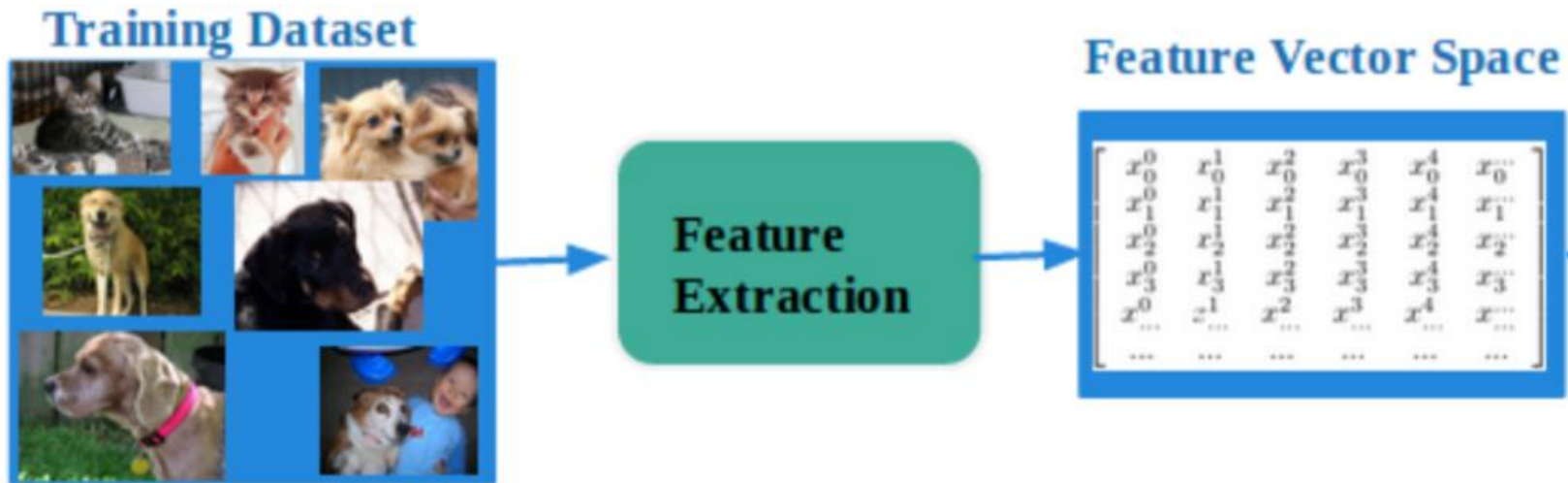


Iris Virginica

- **Feature Extraction:**

SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3	1.4	0.2	Iris-setosa
7	3.2	4.7	1.4	Iris-versicolor
6.4	3.2	4.5	1.5	Iris-versicolor
6.3	3.3	6	2.5	Iris-virginica
5.8	2.7	5.1	1.9	Iris-virginica

Feature Extraction (Cont..)



- Gray level co-occurrence matrix (GLCM)
- Histogram of oriented gradients (HOG)
- Speeded-up robust features (SURF)
- Local binary patterns (LBP)
- Haar wavelets
- Color histograms

Data Preprocessing

- **Normalization (Scaling):**

$$X_{\text{new}} = \frac{X_i - \min(X)}{\max(x) - \min(X)}$$

- **One Hot Encoding:**
- Encode categorical labels as one-hot numeric array.

Human-Readable

Machine-Readable

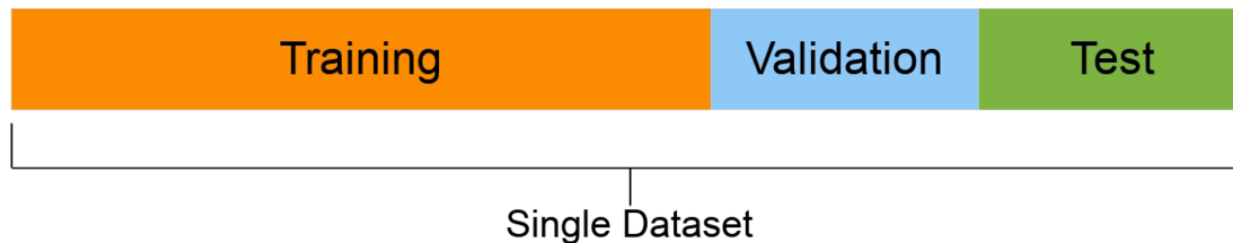
Pet	Cat	Dog	Turtle	Fish
Cat	1	0	0	0
Dog	0	1	0	0
Turtle	0	0	1	0
Fish	0	0	0	1
Cat	1	0	0	0

Feature Selection

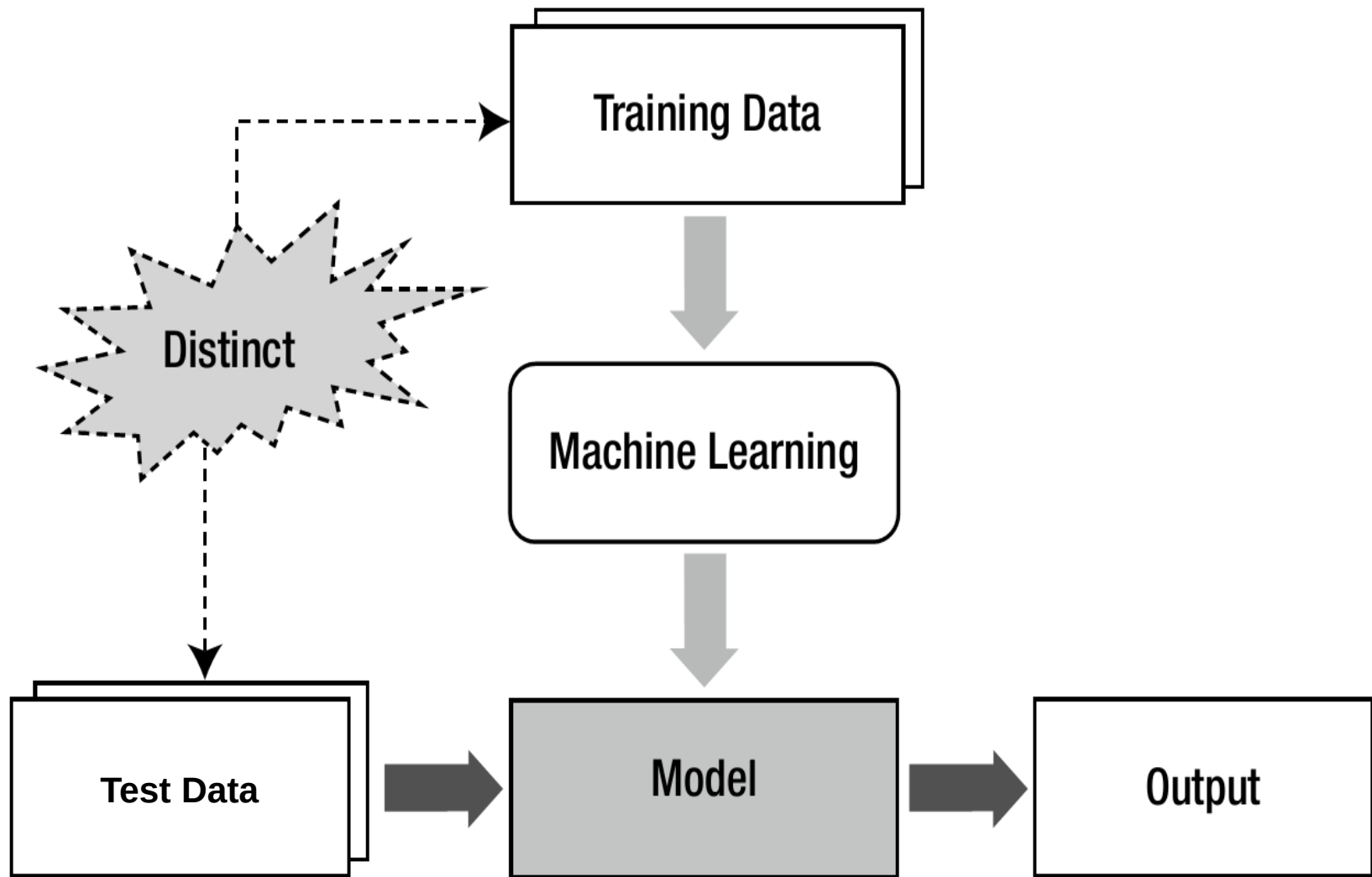
- Feature selection is the process of reducing the number of input variables (features).
- Also known as dimensionality reduction.
- **Why to go for feature selection?**
 - To reduce the computational cost
 - In some cases, to improve the performance of the model.
- **Feature Selection Techniques:**
 - Principal Component Analysis
 - Fisher discriminant analysis
 - Information Gain

Training-Validation-Testing

- The total dataset is randomly split into three sets:
 - Training set (typically 60% of the dataset),
 - Validation set (typically 20%) and
 - Test set (the remaining 20%).



- During training, we train using only the training data. But during training, we evaluate the loss and accuracy of the model on validation data.
- **Model Evaluation**
- When training is finished, we evaluate the model loss and accuracy on the test data.



Machine Learning Flow

(Steps used in Machine Learning)

- 1. Collecting data:** Be it the raw data from excel, text files etc., this step (gathering past data) forms the foundation of the future learning.
 - The better the variety, density and volume of relevant data, better the learning prospects for the machine becomes.
- 2. Preparing the data:** Any analytical process thrives on the quality of the data used.
 - One needs to spend time determining the quality of data and then taking steps for fixing issues such as missing data and treatment of outliers.
- 3. Training a model:** This step involves choosing the appropriate algorithm and representation of data in the form of the model.
 - The cleaned data is split into two parts – train and test (proportion depending on the prerequisites);
 - the first part (training data) is used for developing the model.
 - The second part (test data), is used as a reference.

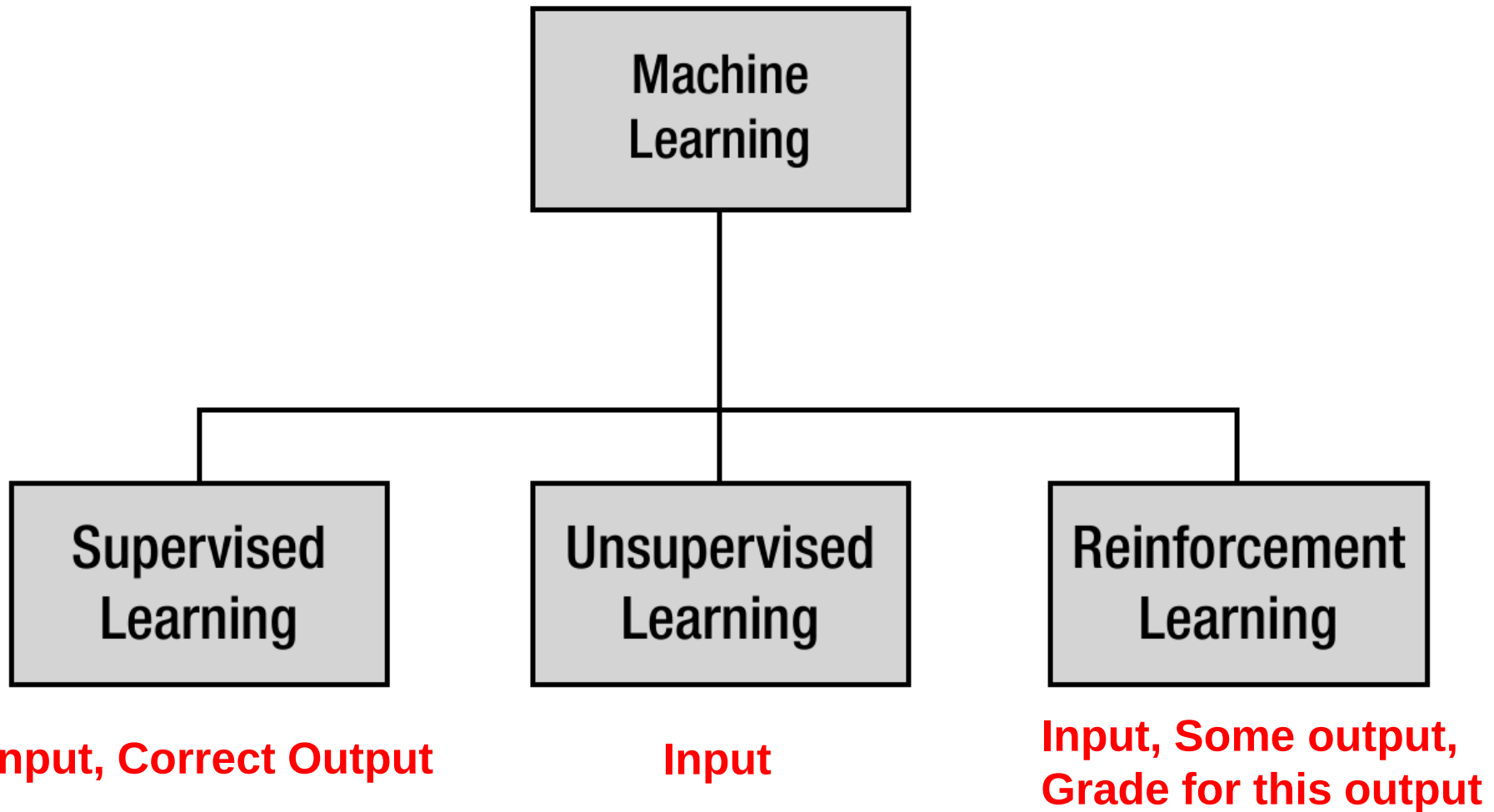
4. Evaluating the model: To test the accuracy, the second part of the data (holdout / test data) is used.

- This step determines the precision in the choice of the algorithm based on the outcome.
- A better test to check accuracy of model is to see its performance on data which was not used at all during model build.

5. Improving the performance: This step might involve choosing a different model altogether or introducing more variables to augment the efficiency.

- That's why significant amount of time needs to be spent in data collection and preparation.

Types of Machine Learning



Supervised Learning

- Supervised learning is a machine learning method in which **models are trained using labeled data.**
- Here, it is assumed that the correct “target” output values are known for each input pattern.
- Supervised learning needs supervision to train the model, which is similar to as a student learns things in the presence of a teacher.
- Supervised learning can be used for two types of problems:
 - **Classification**
 - **Regression**

Unsupervised Learning

- Unsupervised learning is another machine learning method in which patterns inferred from the **unlabeled input data**.
- The goal of unsupervised learning is to find the patterns from the input data.
- Unsupervised learning does not need any supervision. Instead, it finds patterns from the data by its own.
- There is no label or target given for the examples.
- Unsupervised learning can be used for:
 - **Clustering and Association Learning**

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.

Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

Reinforcement Learning

- Reinforcement Learning is a feedback-based Machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions.
- For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.
- Consider an example of a child learning to walk.



- Let's formalize the above example, the “problem statement” of the example is to walk, where the child is an agent trying to manipulate the environment (which is the surface on which he walks) by taking actions (viz walking) and he tries to go from one step to another.
- The child gets a reward (let's say chocolate) when he accomplishes a submodule of the task (viz taking couple of steps) and will not receive any chocolate (a.k.a negative reward or penalty) when he is not able to walk.
- This is a simplified description of a reinforcement learning problem.

Semi-Supervised Learning

- Problems where you have a large amount of input data (X) and only some of the data is labeled (Y) are called semi-supervised learning problems.
- Many real world machine learning problems fall into this area.
- This is because it can be expensive or time-consuming to label data as it may require access to domain experts, whereas, unlabeled data is cheap and easy to collect and store.
- One can also use supervised learning techniques to make best guess predictions for the unlabeled data, feed that data back into the supervised learning algorithm as training data and use the model to make predictions on new unseen data.

Classification Regression and Clustering

- Classification and Regression are supervised learning approach, while clustering is a unsupervised learning approach.
- **Classification:**
 - In classification, the data is categorized under different labels (classes).
 - Here, we want to know which class the input belongs to.
 - **Output is categorical (discrete value).**
 - Examples:
 - **Spam mail filtering:** Classifies the mails in regular or spam.
 - **Digit recognition service:** Classifies the digit image into 0 to 9.
 - **Face recognition service:** Classifies the face image into one of the registered users.

- **Regression:**

- The Regression analysis is the statistical model which is used to predict the numeric data instead of labels.
- **Output is continuous value.**

- **Examples:**

- Estimating income by age.
- Estimating price of a house by its size.
- Estimating price of a car by its model and purchase year.
- Estimating share prices for market analysis.

- **Clustering:**

- Clustering is the task of partitioning the dataset into groups, called clusters.
- The goal is to split up the data in such a way that points within single cluster are very similar and points in different clusters are different.
- It determines grouping among unlabeled data.

